Feature	MQTT	CoAP	OPC UA
Standard body	OASIS (also ISO/IEC 20922)	IETF (RFC 7252)	OPC Foundation (IEC 62541)
Model	Publish/subscribe via broker	Request/response, like HTTP	Client/server and publish/subscribe
Transport	TCP	UDP	TCP, or MQTT/UDP for pub/sub
Device size	Small to large	Very small	Medium to large
Data meaning	Not defined	Not defined	Rich information model
Best for	Telemetry, cloud messaging	Battery sensors, mesh networks	Factories, machines
Open source tools	Mosquitto, Paho, EMQX	libcoap, Californium, aiocoap	open62541, Milo, asyncua

Table 1: A comparison of three key open standards

Eclipse Mosquitto is an open source broker that supports it.

To date, open standards most at the heart of IoT are MQTT, CoAP and OPC UA. Let's take a closer look at each.

MQTT: The common messenger

Message Queuing Telemetry Transport (MQTT) is a lightweight messaging protocol. It was developed in 1999 to track pipelines over slow satellite communications. Nowadays, it's an OASIS standard. Version 3.1.1 has been published as ISO/IEC 20922. The current major version is 5.0 (MQTT).

MQTT is a publish/subscribe protocol. The devices are not directly connected. Instead, these communicate with a central server known as a broker.

- A sensor sends a message to a topic, e.g., farm/field1/soil-moisture.
- All apps that want this data subscribe to the same topic.
- Each message is sent by the broker to all the subscribers.

This is the simple model that is the key to interoperability. The sensor need not be aware of who accesses its data. In the application, it is not important to know which vendor made the sensor. Both only need to know the name of the topic and the format of the message.

MQTT is also very lightweight. The smallest message header is just 2 bytes. It is based on TCP and is good for slow or unreliable networks. It has three Quality of Service (QoS) levels:

- *QoS 0 (at most once):* Quick, but a message might be lost.
- *QoS 1 (at least once):* The message is delivered but can be delivered twice.
- *QoS 2 (exactly once):* Safest, but slowest.

Open source support: Eclipse Mosquitto, EMQX and VerneMQ are widely-used open source brokers. There are client libraries for Eclipse Paho in C, Python, Java, JavaScript, and more.

CoAP: The web for tiny devices

CoAP is the acronym for 'Constrained Application Protocol'. It is an IETF

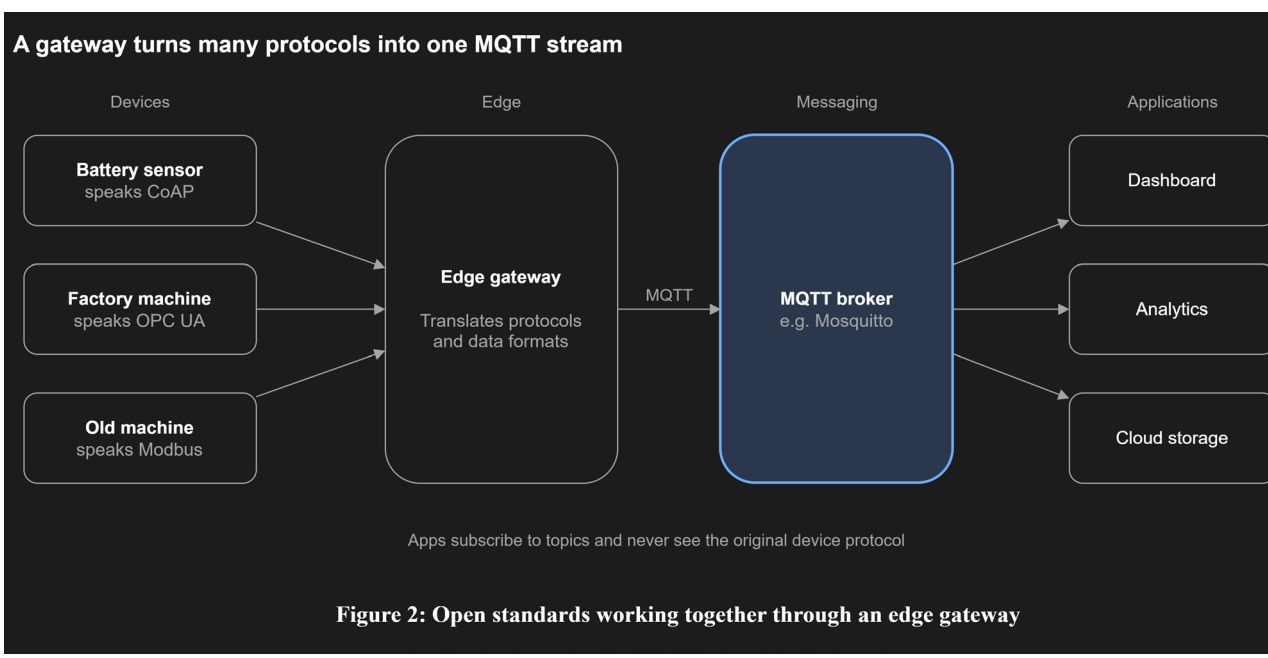


Figure 2: Open standards working together through an edge gateway

Figure 2: Gateway and MQTT broker